HTML (Hyper Text Markup Language)

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→ HTML is used to define layout and attribute of www document as well as to create links b/w web pages.

→ Browser donot display HTML tags, but use them to render content of page

- HTML is subset of **SGML** (Standard Generalised Markup Language).
- Markup language is a way of writing layout information within documents
- developed by Tim Berners Lee at CERN in 1990.

Main feature of HTML

→ Linking : data can be linked in HTML that allows you to be move from one page to another or from top to bottom and viceversa.

→ Simplicity : HTML is simple, easy to use and understand.

→ Portability : It uses ASCII format, HTML is platform independent

HTML Elements and Tags.

(i) HTML Element usually consist of start tag and end tag, with content inserted in between. HTML elements represent semantics or meanings

(ii) HTML tags are element names surrounded by angle brackets.

<tagname> content </tagname>

↓ start tag ↓ end tag (forward slash sign)

(iii) All HTML elements can have attributes (providing additional information) about an element. They are always specified in start tag. ex: <img src="....." width="500">

↓ attribute ↓ attribute.

There are two kinds of element

(i) Container Element:

↳ with start and end tag.

eg: <html> </html>

(ii) Empty Element (standalone tags)

↳ only with start tag, there is no closing tag.

eg: <hr>

HTML TAG STRUCTURE

<Tag_Name Attribute_Name = "value"> content </Tag-Name>

quotes are optional

eg: ....

<body bgcolor = "red"> ... </body>

Simple Example Explained.

<! DOCTYPE html> → defines document to HTML 5

<html> → root element of HTML page

<head> → meta info about document

<title> Page title </title> → specifies title of document

</head>

<body> <h1> My first heading </h1> → defines large heading

 My first para → defines paragraph.

</body>

</html> → indicates end of webpage.

<body> ↓ contains visible page content

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Basic HTML Tags.

(i) HTML

<html> element defines whole document.
represent root element of HTML document.
it informs web browser document is webpage
Syntax: <html>
⋮
</html>

attributes → dir (specifies type of text to be presented by browser. By default: "ltr" ie left to right. The "rtl" is used for language like chinese etc.

→ lang (specifies language of text)

eg: <html dir = "ltr" lang = "en" > ... </html>
↓
english

(ii) <head>
Container for all header element.
↳ provides info to browser and searchengine but not displayed to person viewing webpage.
↳ includes <title> for document
Syntax: <head> ... </head>

(iii) <title>
defines title of document
Syntax: <head>
<title> ... </title>
</head>

(IV) <body>

defines document's body

contains all contents of an HTML document, such as text, hyperlinks, images, tables, lists etc.

Syntax:

```
<body>
  :
</body>
```

attributes → background: specifies background image for a document

→ text: specifies text color. By default: black.

→ bgcolor: specifies background color of document.

→ link: unvisited link color. By default: blue
alink: active link color. By default: red
vlink: visited link color. By default: purple.

→ topmargin: sets top margin (b/w top of document and top of browser window)
leftmargin: left hand margin (b/w left side of document and left edge of browser window)

Syntax:

```
<body background="URL of image.jpg" text="color name/hex_number/rgb_number" bgcolor="color name" link="color name" topmargin="value" leftmargin="value"> ... content ...
</body>
```

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eg: <body background = "flower.jpg" bgcolor = "pink"
leftmargin = "10" topmargin = "100" text = "blue" >
content </body>

(v)
tag specifies font style, size, color of text
used to change appearance of short segment of text.

attributes → size: specifies size of text. 1 is smallest and 7 is largest.

face: specifies font name or type of text

color: specifies color of text.

eg: < font face = "Times New Roman" size = 6 color = "red">
content

we can set relative size to current size using (+)ve or (-ve) sign. eg: size = +2.

(vi) <basefont>
↳ this specifies default font color, font size, font face for all text in document that follows it.
↳ only supported by Internet Explorer web browser.
Ending </basefont> tag is optional.

attributes → color, size, face. (same as above)

eg: < basefont face = "Times New Roman" size = 5 color = "red">
content

Some important color and their hexadecimal and RGB values.

	Color Name	Hexadecimal value	RGB value
(i)	Black	#000000	rgb(0,0,0)
(ii)	White	#FFFFFF	rgb(255,255,255)
(iii)	Red	#FF0000	rgb(255,0,0)
(iv)	Yellow	#FFFF00	rgb(255,255,0)
(v)	Blue	#0000FF	rgb(0,0,255)

(7)
is used to centralise a segment of text to be displayed on browser's window.
Syntax:
 ...

(8) Comment tag <!-- ... --> tag
used to insert comments in source code of web pages.
syntax: <!-- ... -->

You use comments to explain your code, it can help you, when you edit source code at later date.

(9) Heading tags.
used to display the text as some heading.
<h1> to <h6>. syntax: <h1> ... </h1>
↳ largest ↳ smallest

attribute → align: left, right, center, justify
eg: <h1 align = "value"> or <h1 align = "left">

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(10) Paragraph tag.
↳ seperate text by paragraph
↳ by default, all text in HTML document is treated as one long paragraph.
↳ tag ends a line and inserts a blank line before next paragraph.

Syntax:

 ...

attributes → align: left, right, center, justify

eg: content

(11) Line break
 (empty element)
↳ inserts a line break ie will be displayed in next line.

content
 or
 content

12) Horizontal Rule <hr> tag. (section seperator)
↳ seperates two section by forming rule or line

attribute → align: specifies alignment of horizontal line center, left, right

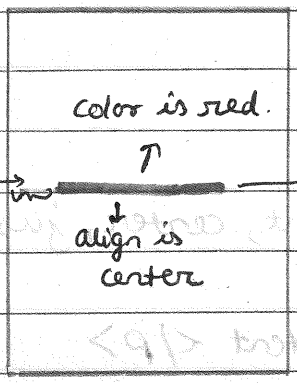
size: specifies height of line. default size is 3 pixels. (in pixels)

noshade: line is drawn with solid black line as 2D effect instead of default 3D effect ie. without shading.

width: specifies width of line (by default 100%)

color: specifies color of horizontal line

eg: <hr align = "center" size = "5" noshade width = "50%" color = "red">



(13) <bgsound> tag.
used to play soundtrack in background.
used only in Internet Explorer only.

syntax:
<bgsound src = "soundurl" loop = "value">

attributes → src: URL (specifies path of sound file.

loop: replay a background music certain number of times.
eg: 5 → repeats song 5 times.

(14) <embed> tag.
- used to designate an area for interactive content (plug in) or an external application.
- useful when you want to add content that might not be natively supported by Internal Explorer.

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eg: <embed src = "....mp3" > </embed>

Text formatting tags.

(i) <q> tag : defines short quotation tag ie. applies double quotes on text.

eg: I am 9 year old. <q> I am not a teenager </q>

Output: I am 9 year old. "I am not a teenager".

(ii) tag : used to add styles to content or manipulate content with for example, Javascript.

eg: My mother has black eyes.

Output: My mother has black eyes.
↳ gives color to text i.e style.

(iii) tag . : used to text display in bold format.

eg:
Hello! How R you.

Output: Hello! **How R you.**
↳ bold.

(iv) <sub> and <sup> tag.
↳ subscript ↳ superscript

eg: H ₂ O a ²
Output: H_2O : a^2

(v) <blockquote> tag

↳ defines long quotation.

↳ browser inserts white space before or after a blockquote element.

eg: I am flower.

<blockquote>

I am pink in color.

I love blooming.

I hate winter.

</blockquote>

Output: I am flower

I am pink in color. I love blooming. I hate winters.

(vi) <del> tag.

↳ defines deleted text.

eg: I belong to <del> Himachal </del> Delhi.

Output: I belong to ~~Himachal~~ Delhi.

(vii) <strike> tag

↳ displayed with ~~strike through~~.

eg: KHUSHBOO Singhal <strike> Publication </strike>

output: KHUSHBOO Singhal ~~Publication~~,

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(viii) <big> and <small> tag.
↳ larger font ↳ smaller font

eg: I have a <big> flower </big> for <small> everyone </small>

Output: I have a flower for you
↳ big ↳ small

(ix) <pre> tag.
↳ used for preformatted text. ie shows text same as you write.

eg: <pre>
I beleive in god
as we are
the creation of him and

we can live...
</pre>

Output: I beleive in god.
as we are
the creation of him and

we can live...

without <pre> tag
output: I beleive in god as we are the creation of him and we can live...

(x) <address> tag.
↳ used for address. Usually in italic.

eg:
```
<address>
    1787 Saket <br>
    filmcenter, Anupan <br>
    New Delhi
</address>
```

Output: 1787 Saket
filmcenter, Anupam
NewDelhi
} italic

Logical text styles. (idiomatic style tag)
↳ gives different look to text when it is displayed by browser.

(i) citation tag → <cite>
used to display title of book & name of character.

(ii) code tage → <code>
used to indicate code of text.

(iii) Emphasis tag →
used for emphasising phrases (in italic)

(iv) Keyboard → <kbd>
defines text that should be entered by user.

(v) Sample → <samp>
indicates sequence of literal character

(vi) Variable → <var>
is used for variables.

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Physical text styles.
Changes physical appearance of text in your webpage.

(I) Bold tag
used to make text in bold form. ...

(II) Italic tag
used to make text in italic form ...

(III) Underline tag <u>
used to underline the text <u>...</u>

(IV) Teletype tag <tt>
Create teletype or monospaced font of enclosed text
<tt>...</tt>

→ Universal Resource Identifier (URI)
is a string of characters used to identify names or resources on Internet.

(I) Fragment Identifiers : Some URI's refer to a location within a resource. This kind of URI ends with "#" followed by anchor identifiers.
eg: http://games.com/html/top.html#section
↳ anchor named section.

(II) Relative URI : doesnot contain any naming scheme information. Its path generally refers to resource on same machine as current document.
- may contain relative path components and may contain fragment identifiers.
- are resolved to full URIs using base URI.

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Hyperlinks

- Link text / image to eachother.
- Created by <a> anchor tag.
- <a> ... ie container tag.

attributes → href : specify url of target document

title : specify title of linked document

name : specify name of anchor.
(to create bookmark inside a document)

eg: <a href = "http://www.facebook.com"
name = "Facebook" title = "hyperlink created">
Content

There are two types of linking

(i) Internal linking : linking within same document
(# symbol is used)

eg: content

 content

(ii) External linking : linking to another document.

 content

Images as links : images can be used as hyperlinks.

 <img src = "image.jpg"> ...

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→ HTML List

are of three types

(i) Unordered list

L unnumbered list or bulleted list

L starts with <ul> tag.

↳ each item listed starts with <li> tag.

↳ ends with tag.

attribute → type (change bullet symbol) : circle (○), disc (●), square (■)

eg :
```
<ul type = "disc">
    <li> apple </li>
    <li> mango </li>
    <li> orange </li>
</ul>
```

not necessary to use tag. ie. optional.

output :
- apple
- mango
- orange

(ii) Ordered list

L numbered list

L starts with <ol>, ends with tag.

L each item listed starts with <li> tag.

attribute → type : "value" can be I, i, 1, A or a

start (change beginning of list)

eg:
```
<ol type = "A"  start = "5" >
<li> Games </li>
<li> study </li>
<li> Eat </li>
</ol>
```

Start from here ↓

A B C D E
1 2 3 4 5

Output :
E Games
F Study
G Eat

 is optional.

(iii) Definition List <dl>
↳ that has description of each item

Syntax:

<dl> data list start
<dt> data term (item)
<dd> definition of data (item)
</dl> data list end.

eg:

```
<dl>
    <dt> apple
    <dd> It is very important to eat apple.
    <dt> oranje
    <dd> you should drink oranje juice
    <dt> mayo
    <dd> ripe mayo are used to make shake.
    </dt>
```

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Output:

apple
It is very important to eat apple.
Orange
you should drink orange juice
Mango
ripe mango are used to make shake.

we can also have nested list

```
<ul>
    <li> Fruits
        <ol type="i">
            <li> mango
            <li> Apple
        </ol>
    <li> Vegetables
        <ol type="i">
            <li> onion
            <li> Tomato
        </ol>
</ul>
```

output
- Fruits
  i. Mango
  ii. Apple
- Vegetable
  i. onion
  ii. Tomato

→ HTML Table

allows you to arrange data in form of rows & columns. data can be in any form whether it may be text, tables, images etc

Tags used are:

(I) <table> ... </table> → table tag

(II) Row tag <tr> ... </tr>

(III) Table heading <th> ... </th>

(IV) Cell data tag <td> ... </td>

(V) Caption tag <caption> ... </caption>

Attributes → border: used to display

bordercolor: defines table border color

frame: which portion of border should be displayed (values = void, above, below, hside, vside, box, border etc)

very important ←

cellspacing: amount of space b/w cells.

cell padding: amount of space b/w content of cell and cell wall. by default, cellpadding is 1.

align: is used to align table completly.

bgcolor: used to specify any desired background color

height & width: specify height & width of table.

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<tr> tag : table row. It is used to create a new row of table.

attributes → align : content in table row (left, right, center, justify)

bgcolor : used to specify a background color for table row

valign : specifies vertical alignment of content in table row (top, middle, bottom, baseline)

<th> tag : denotes table header. data inserted appears in bold. (apart from this, <th> tag and <td> tag are similar)

attributes → align, bgcolor, valign. (same as above)

colspan : specifies number of column ie allow user to strech a cell to span multiple columns.

rowspan : used to span multiple rows.

height & width : specifies height & width of cell.

<td> tag : denotes table data.

attributes : (same as above)

<caption> tag

<caption> tag is used to define table caption.

attribute → align : left / right / top / bottom / center

```
eg: <table align = "center" border = "4"
    bordercolor = "blue" cellspacing = "5" cellpadding =
    "4" height = 20% width = 20%. bgcolor = "pink" >

    <tr align = "right" bgcolor = "skyblue"
      valign = "middle" >

    <th align = "center" colspan = 2 bgcolor = "white" >
      class </th>

    <th rowspan = 2 > VIII </th>
    </tr>

    <tr>
    <td> A </td>
    <td> B </td>
    </tr>

    <tr>
    <td> C </td>
    <td> D </td>
    </tr>
    <caption align = "center"> class info </caption>

</table>
```

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because of colspan

Class		→ bold because of <th> tag
VIII	A	→ tr
	B	→ td
C	D	

bcz of rowspan

class info → caption

→ HTML Form

L usually implemented to collect data from user and use it in webpages.

L information collected can be written to a file, submitted to database, E-mail to someone.

It contains various attributes

(i) Name : specifies name of form. Name is optional if there is only one form on webpage

(ii) Action : provides URL of program that receives info from form and processes it.

Get URL is limited in size.

(iii) Method : used to submit data of form. (non secure)

L can take three values : data will be visible in address field.

(a) GET : sends information to server. (by default)

(b) POST : sends form information in HTTP environment

(c) Form : opens a new form as per URL specified in action attribute.

used to send large amount of data. (for sensitive & personal information)

Syntax :

```
<form name = "form_name" action = "URL"
    method = "method">
```

In GET user can bookmark result, In POST user cannot bookmark

<input> tag

- defines text entry fields, checkboxes, radio buttons, push buttons.
- it is empty tag

attributes → **type** : display type of field (text, password, radio, checkbox, submit, reset button, image, hidden)

name : specifies fieldname. Required for identification.

value : specifies default value of input tag.

eg: <input type = "text" name = "first_name" value = "Mickey" >

size : resize field such as set length of text box.

height & width : set height & field of text.

max length : allows to specify max no. of character a user can enter

align : align content

(i) Text — Hello

(ii) Password — ****

(iii) Radio — ○ Male, ● Female } only one option at a time.

(iv) Checkbox — ☑ Undergraduate, ☑ Postgraduate, ☐ Matrix } many option at a time.

additional attribute. → submit : provides button on form to submit data.

reset : provides button on form to reset input fields.

hidden : used to hide data inside page which later on can be pushed to server.

image : defines image as submit button.

<select> tag
- used to create drop-down menus and scrolled lists
- it is container tag.

attributes → name : shows name of drop-down

size : displays height of list. If it is 1 (or omitted), list is drop-down menu, otherwise scrolled list

disabled : specifies drop-down list is disabled.

<legend> : elements defines a caption for <fieldset> element.

<fieldset> : element is used to group related data in a form.

eg:
```
<form action = "/action-page.php">

<fieldset>
<legend> Information </legend>

Firstname : <br>
<input type = "text" name = "first name" value =
"Mickey"> <br>

<input type = "submit" value = "Submit">

</fieldset>
</form>
```

output:

